

PLSC/DATA 40501: Machine Learning in the Social Sciences

FALL 2026

Time and Location	T 2:00 - 4:50 PM, Harper 130
Course Credit	100 units
Professor	Isaac Mehlhaff (imehlhaff@uchicago.edu)
Office Hours	Th 2:00 - 4:00 PM, Pick 413 (calendly.com/imehlhaff/office)

Course Description and Objectives

This course offers an applied introduction to machine learning for social science research, focusing on how models can be used to extract patterns from complex (and often unstructured) data, generate predictions, and assess the quality of those predictions. We will explore both the logic and practical implementation of key machine learning methods, including tree-based models, neural networks and deep learning, kernel-based models, and clustering techniques.

Each class period will be divided into approximately two halves, with a short break in between. The first half will be dedicated to lecture and discussion about the topic for that week, while the second half will be a lab session focused on implementing the methods covered that week, generally with applied social scientific data. Beginning in Week 3, students will be responsible for leading discussions and designing lab exercises. Accordingly, the order and content of the course will largely depend on student interests. The course prioritizes computational skill-building and is suitable for graduate students in the social sciences, data science, or digital humanities.

Prerequisites

Students must enter the course with a general background in statistical programming and inference. Experience with common mathematical operations in calculus and linear algebra is preferred, as is a working knowledge of Python.

Textbooks

We will draw primarily from two textbooks. Both are freely available online if you do not wish to purchase them:

- Christopher M. Bishop. *Pattern Recognition and Machine Learning*. New York: Springer (2006). <https://www.microsoft.com/en-us/research/wp-content/uploads/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf>.
- Kevin P. Murphy. *Probabilistic Machine Learning: An Introduction*. Cambridge, MA: The MIT Press (2021). <https://github.com/probml/pml-book/releases/latest/download/book1.pdf>.

Bishop and Murphy tend to occupy a middle ground, presenting high-level treatments of the material without being excessively dense. If you desire more mathematical detail or more accessible explanations, I recommend consulting Hastie et al. or James et al., respectively:

Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. New York: Springer (2009). https://hastie.su.domains/ElemStatLearn/printings/ESLII_print12_toc.pdf.download.html.

Gareth James, Daniela Whitten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor. *An Introduction to Statistical Learning: with Applications in Python*. New York: Springer (2023). <https://www.statlearning.com>.

Finally, if you find yourself in need of more practical, hands-on guidance for implementing machine learning techniques in Python, I recommend the following resources:

François Chollet. *Deep Learning with Python*. 2nd edition. Shelter Island, NY: Manning Publications (2021). <https://sourestdeeds.github.io/pdf/Deep%20Learning%20with%20Python.pdf>.

Aurélien Géron. *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. 2nd edition. Sebastopol, CA: O'Reilly Media (2019). http://14.139.161.31/OddSem-0822-1122/Hands-On_Machine_Learning_with_Scikit-Learn-Keras-and-TensorFlow-2nd-Edition-Aurelien-Geron.pdf.

Wes McKinney. *Python for Data Analysis*. 3rd edition. Sebastopol, CA: O'Reilly Media (2022). <https://wesmckinney.com/book/>.

Software

Before our first class meeting, you must have all the following software resources installed on your machine, and you must have ensured they are all operational.

- **Python:** Python is the most widely used, most versatile, and best-supported open-source programming language for computational work. The easiest method to install Python is via Miniconda: <https://docs.conda.io/projects/conda/en/latest/user-guide/install/index.html>.
- **Conda:** Regardless of how you install Python, please ensure you have a working installation of conda: <https://docs.conda.io/projects/conda/en/latest/user-guide/install/index.html>. Conda provides a convenient infrastructure for managing Python library installations, and we will use it extensively this quarter.
- **Virtual Studio Code (optional):** Python code can be written in any plain text editor, and you can choose the workflow that is best for you. However, I strongly recommend using Virtual Studio Code (VS Code): <https://code.visualstudio.com/>. VS Code provides a highly functional development environment for writing, running, and debugging Python code. It also connects seamlessly with git, GitHub, and AI agents. This quick start guide can help you

get started with Python in VS Code, but you should explore it yourself and customize it to your liking: <https://code.visualstudio.com/docs/python/python-quick-start>.

- **Git:** We will use git and GitHub for version control and to organize our course materials. If you have a Mac, I recommend installing git via Homebrew. If you have a Windows machine, I recommend installing git via Winget. Instructions can be found here: <https://git-scm.com/install/windows>.
- **GitHub:** Ensure you have an active GitHub account prior to the first class meeting: <https://github.com/>. A quick start guide to git and GitHub can be found here: <https://docs.github.com/en/get-started/start-your-journey/git-and-github-learning-resources>. GitHub Desktop can also be a handy app for managing repositories locally, but this is optional: <https://github.com/apps/desktop>.

Grading and Logistics

Graduate school is about learning and developing skills; getting good grades is not the objective. Nevertheless, course grades will be based on the following categories. All assignments are due at 11:59 PM on the indicated dates.

- **Participation (10%):** Practicing how to engage on scholarly matters is an important part of your professional development. As this is a seminar-style class, you are expected to come prepared with questions and actively engage in class discussion. This can include asking conceptual, technical, or troubleshooting questions on Slack and helping respond to your colleagues' inquiries. Being engaged in class also entails not allowing yourself to be distracted by technology, including AI tools.
- **Lab assignments (15%):** The second half of our class meetings will consist of you working individually or in small groups to complete a Python script applying concepts from lecture and readings. Even though you may collaborate with others, you must submit your own assignment. Lab assignments will be due on Friday of the week they are assigned and will be graded on completion. Submit by pushing to your remote GitHub branch.
- **DataCamp assignments (15%):** To complement the more conceptual training we conduct in class (and to ensure you understand how models are implemented without the aid of AI), you will complete coding exercises on DataCamp: <https://app.datacamp.com>. DataCamp assignments will be due the Monday before each concept is scheduled to be discussed in class and will be graded on completion.
- **Leading discussion (30%):** Starting in Week 3, you will work in small groups to lead class sessions, with each student responsible for one session throughout the quarter. Groups should prepare a lecture to guide discussion and a lab exercise to demonstrate that week's method. The lecture should detail the following:
 1. The goal of the method.
 2. The intuition behind how and why it works.

3. Mathematical details (e.g. derivations and proofs) underpinning the method.
4. High-level presentations of algorithms.
5. Benefits and possible areas of complication when using the method.
6. Approaches for implementing the method in software.
7. Applied examples in political or other social sciences.
8. Any other information germane to understanding the method.

Practice good pedagogy: use illustrations, examples, diagrams, and live mathematical and coding exercises, where appropriate. Groups *must* meet with me to discuss content and get feedback on their plans for the lecture and lab. Discussion leaders will be graded on the degree to which they accurately, thoroughly, and effectively present material. Distribute materials to the class by making a pull request to our main GitHub branch.

- **Final project (30 %):** You will conduct a final project that uses machine learning methods to solve a problem in your area of research. The scope of what would constitute an acceptable project is very wide; please consult with me if you have questions about whether your idea is appropriate. Regardless of the topic, your project will take the form of a scientific poster (size A10), similar to what would be presented at PolMeth, for example. The virtual poster will be due on December 11 and will be graded on the degree to which 1) the problem is clearly stated, 2) the proposed solution is compelling and well-justified, 3) the method is thoroughly evaluated, and 4) the results are clearly communicated. Remember: tables, figures, and mathematical expressions tend to communicate more information than words. There are many $\text{\LaTeX}$ templates available online.

A final percentage grade x will translate into letter grades as follows:

- $90 \leq x \rightarrow \text{A}$
- $80 \leq x < 90 \rightarrow \text{B}$
- $70 \leq x < 80 \rightarrow \text{C}$
- $60 \leq x < 70 \rightarrow \text{D}$
- $x < 60 \rightarrow \text{F}$

Course Schedule

All readings—other than those in the textbooks listed above—will be provided in the course GitHub repository. If you find any value in having physical copies or tend to get distracted by technology, I encourage you to print out the readings or purchase copies of the books.

If you have limited experience with the software tools we will use in this course, I strongly recommend working through the following optional DataCamp assignments, ideally before our first class meeting: Introduction to Shell, Introduction to Git, Intermediate Git, Python for R Users, and Intermediate Python.

Week 1 (9/29): Foundational Tools and Concepts

Assignments

- Lab (10/2)

Readings

- Bishop 1 – 1.1, 1.2.4 – 1.2.5, 1.4 – 1.5.2
- Murphy 1.1 – 1.4 (skip 1.3.4), 1.6, 4.1 – 4.2.2, 4.3 – 4.3.1, 8.1 (skip 8.1.2, 8.1.3.3 – 8.1.3.4, 8.1.4.1)
- Cranmer, Skyler J. “Introduction to the Virtual Issue: Machine Learning in Political Science.” *Political Analysis* (n.d.).

Week 2 (10/6): Model Optimization, Evaluation, and Workflow

Assignments

- DataCamp (10/5): Python Toolbox 1 – 2, Supervised Learning with `scikit-learn`
- Lab (10/9)

Readings

- Bishop 1.3, 3.2
- Murphy 4.5.4 – 4.5.7, 4.7.6, 5.1, 5.3 – 5.3.1.1, 5.4 – 5.4.1.1, 5.4.3, 8.1.4 – 8.3.1, 8.4 – 8.4.2
- Grimmer, Justin, Margaret E. Roberts, and Brandon M. Stewart. “Machine Learning for Social Science: An Agnostic Approach.” *Annual Review of Political Science* 24 (2021): 395-419.

Week 3 (10/13): Linear Models and Regularization

Assignments

- DataCamp (10/12): Linear Classifiers in Python 1 – 3
- Lab (10/16)

Readings

- Bishop 3 – 3.1.4, 3.6, 4 – 4.1.4, 4.3 – 4.3.4
- Murphy 4.5 – 4.5.3, 9 – 9.2.3, 9.3 – 9.4 (skip 9.3.2 – 9.3.3), 11.1 – 11.5 (skip 11.2.3 – 11.2.4)

- Blair, Robert A., Christopher Blattman, and Alexandra Hartman. “Predicting Local Violence: Evidence from a Panel Survey in Liberia.” *Journal of Peace Research* 54, no. 2 (March 2017): 298-312.
- Yaman, Selim, Yasir Atalan, and Jeff Gill, “Bridging Prediction and Theory: Introducing the Bayesian Partially-Protected Lasso.” *Electoral Studies* 87 (February 2024): 102730.

Week 4 (10/20): Tree-Based Methods

Assignments

- DataCamp (10/19): Machine Learning with Tree-Based Models in Python
- Lab (10/23)

Readings

- Bishop 14.2 – 14.4 (skip 14.3.1 – 14.3.2)
- Murphy 18 (skip 18.5.3 – 18.5.4)
- Montgomery, Jacob M. and Santiago Olivella. “Tree-Based Models for Political Science Data.” *American Journal of Political Science* 62, no. 3 (July 2018): 729-744.
- Montgomery, Jacob M., Santiago Olivella, Joshua D. Potter, and Brian F. Crisp. “An Informed Forensics Approach to Detecting Vote Irregularities.” *Political Analysis* 23, no. 4 (2015): 488-505.

Week 5 (10/27): Neural Networks and Deep Learning

Assignments

- DataCamp (10/26): Introduction to Deep Learning with Keras
- Lab (10/30)

Readings

- Bishop 5 – 5.3.3, 5.4 – 5.4.1, 5.5 – 5.5.3
- Murphy 13.1 – 13.2, 13.4 – 13.5 (skip 13.4.5)
- Lall, Ranjit and Thomas Robinson. “The MIDAS Touch: Accurate and Scalable Missing-Data Imputation with Deep Learning.” *Political Analysis* 30, no. 2 (April 2022): 179-196.
- Muchlinski, David, Xiao Yang, Sarah Birch, Craig Macdonald, and Iadh Ounis. “We Need to Go Deeper: Measuring Electoral Violence Using Convolutional Neural Networks and Social Media.” *Political Science Research and Methods* 9, no. 1 (January 2021): 122-139.

Week 6 (11/3): Dimensionality Reduction and Clustering

Assignments

- DataCamp (11/2): Unsupervised Learning in Python 1 – 3
- Lab (11/6)

Readings

- Bishop 2.3.9, 9 – 9.2, 9.3.2, 12 – 12.2 (skip 12.2.1 – 12.2.3)
- Murphy 1.3.4, 3.5, 7.4 – 7.4.3, 7.5 – 7.5.2, 20 – 20.1.1, 20.1.3.3, 20.1.4 – 20.2.1, 20.4 (skip 20.4.4 – 20.4.9), 21.1 – 21.5.3
- Ahlquist, John S. and Christian Breunig. “Model-Based Clustering and Typologies in the Social Sciences.” *Political Analysis* 20, no. 1 (2012): 92-112.
- De La Cerda, Nicolás and Jacob R. Gunderson. “Are Party Families in Europe Ideologically Coherent Today?” *European Journal of Political Research* 63, no. 3 (August 2024): 1208-1226.

Week 7 (11/10): Computer Vision

Assignments

- DataCamp (11/9): Image Processing in Python, Image Modeling with Keras 1 – 3
- Lab (11/13)

Readings

- Bishop 5.5.3 – 5.5.6, 8.3.3
- Murphy 2.8.5, 14.1 – 14.6.1, 19.1 – 19.2
- Joo, Jungseock and Zachary C. Steinert-Threlkeld. “Image as Data: Automated Content Analysis for Visual Presentations of Political Actors and Events.” *Computational Communication Research* 4, no. 1 (February 2022): 11-67.
- Torres, Michelle and Francisco Cantú. “Learning to See: Convolutional Neural Networks for the Analysis of Social Science Data.” *Political Analysis* 30, no. 1 (January 2022): 113-131.

Week 8 (11/17): Natural Language Processing

Assignments

- DataCamp (11/16): Feature Engineering for NLP in Python, Unsupervised Learning in Python 4
- Lab (11/20)

Readings

- Bishop 2.2, 13 – 13.1
- Murphy 1.5.4, 15.1 – 15.2, 20.5
- Rodriguez, Pedro and Arthur Spirling. “Word Embeddings: What Works, What Doesn’t, and How to Tell the Difference for Applied Research.” *The Journal of Politics* 84, no. 1 (January 2022): 101-115.
- Roberts, Margaret E., Brandon M. Stewart, Dustin Tingley, Christopher Lucas, Jetson Leder-Luis, Shana Kushner Gadarian, Bethany Albertson, and David G. Rand. “Structural Topic Models for Open-Ended Survey Responses.” *American Journal of Political Science* 58, no. 4 (October 2014): 1064-1082.

Week 9 (12/1): LLMs for Text Classification

Assignments

- DataCamp (11/30): Introduction to Embeddings with the OpenAI API 1 – 2, Fine-Tuning with Llama 3
- Lab (12/4)

Readings

- Murphy 15.4 – 15.5.5 (skip 15.4.2 – 15.4.3), 15.7, 19.2 – 19.3.2.1, 19.3.6 – 19.4 (skip 19.3.6.3), 19.6 – 19.7
- Bosley, Mitchell, Saki Kuzushima, Ted Enamorado, and Yuki Shiraito. “Improving Probabilistic Models In Text Classification Via Active Learning.” *American Political Science Review* 119, no. 2 (May 2025): 985-1002.
- Egami, Naoki, Musashi Hinck, Brandon M. Stewart, and Hanying Wei. “Using Large Language Model Annotations for the Social Sciences: A General Framework of Using Predicted Variables in Downstream Analyses.” *American Journal of Political Science* (forthcoming).

Other Topics

Kernel Methods

- Bishop 6 – 6.1, 6.4 – 6.4.3, 6.4.5, 7 – 7.1.4 (skip 7.1.1)
- Murphy 16.3 (skip 16.3.4), 17 – 17.2.3, 17.3 (skip 17.3.3 – 17.3.5)
- Bonica, Adam. “Inferring Roll-Call Scores from Campaign Contributions Using Supervised Machine Learning.” *American Journal of Political Science* 62, no. 4 (October 2018): 830-848.

- Hainmueller, Jens and Chad Hazlett. “Kernel Regularized Least Squares: Reducing Misspecification Bias with a Flexible and Interpretable Machine Learning Approach.” *Political Analysis* 22, no. 2 (2014): 143-168.

Online Learning, EM, and Variational Inference

- Bishop 2.3.5, 3.1.3, 9.3 – 9.3.1, 9.4, 10 – 10.1.1
- Murphy 4.4.2, 8.4 – 8.4.3, 8.7 – 8.7.3
- Grimmer, Justin. “An Introduction to Bayesian Inference via Variational Approximations.” *Political Analysis* 19 no. 1 (2011): 32-47.
- Imai, Kosuke, James Lo, and Jonathan Olmsted. “Fast Estimation of Ideal Points with Massive Data.” *American Political Science Review* 110, no. 4 (November 2016): 631-656.

Audio Signal Processing

- Bishop 13.2 – 13.2.5 (skip 13.2.3 – 13.2.4)
- Murphy 15.3
- Dietrich, Bryce J., Matthew Hayes, and Diana Z. O’Brien. “Pitch Perfect: Vocal Pitch and the Emotional Intensity of Congressional Speech.” *American Political Science Review* 113 no. 4 (November 2019): 941-962.
- Knox, Dean and Christopher Lucas. “A Dynamic Model of Speech for the Social Sciences.” *American Political Science Review* 115, no. 2 (May 2021): 649-666.

Graph-Based Methods

- Bishop 8 – 8.1.1, 8.2 – 8.3.1, 8.4 – 8.4.4
- Murphy 3.6, 23.1 – 23.3.1, 23.3.6 – 23.4.3.2, 23.6
- Dorff, Cassy, Max Gallop, and Shahryar Minhas. “[W]hat Lies Beneath: Using Latent Networks to Improve Spatial Predictions.” *International Studies Quarterly* 66, no. 1 (September 2022): sqab086.
- Lo, Adeline, Santiago Olivella, and Kosuke Imai. “A Statistical Model of Bipartite Networks: Application to Cosponsorship in the United States Senate.” *Political Analysis* 34, no. 3 (July 2026): 451-470.

Communication

We will use our course's private Slack channel for asking and answering questions about course material or logistics. Rather than emailing questions to me, please post your question on Slack so other students can also benefit from the discussion. If you are asking a question about coding, it is good practice to provide a minimal working example (MWE) to help others understand your problem. Good examples of how to create an MWE can be found at <https://stackoverflow.com/questions/5963269/how-to-make-a-great-r-reproducible-example> and <https://stackoverflow.com/help/minimal-reproducible-example>.

I am always happy to meet with you during my scheduled office hours or outside those hours if they do not work with your schedule. My office hours are generally first-come, first-served, and I encourage you to attend with classmates if you have similar questions. If you prefer a one-on-one meeting, the best option is to sign up for a time slot using the link at the top of this document.

Learning is a collaborative endeavor, and we all bring unique backgrounds and experiences to the course material. Treat your colleagues how you want to be treated. You will need to collaborate on presentations and labs; use these opportunities to learn from each other. I encourage you to form study groups and discuss material amongst yourselves if you do not understand it.

Academic Integrity

Familiarize yourself with the university's policies on academic dishonesty and plagiarism: <https://studentmanual.uchicago.edu/academic-policies/academic-honesty-plagiarism/>. As with most other things in this profession, you should give credit to others when you use their language, materials, or findings. There could be serious consequences for committing plagiarism, including failing the course and being asked to leave the university, not to mention reputational and career implications.

Generative Artificial Intelligence

Use of generative AI in this course is permitted, provided that any code or text generated by AI is attributed as such. I further encourage you to use these tools as a tutor for concepts that you are confused about and to help flatten the learning curve if you are unfamiliar with Python or other coding languages. But beware: I will grade assignments accordingly. Generative AI frequently makes mistakes, its scientific writing is typically low-quality, and using it as a crutch will hinder your own ability to learn the material. You are solely responsible for all work you submit. I reserve the right to revoke this privilege at any time if I suspect it is being abused.

Attendance, Late Work, and Accommodations

This is a PhD-level course; you are expected to come to class and complete assignments on time. If you need to be absent from class or extenuating circumstances prevent you from completing assignments, please alert me ahead of time. I will not accept late assignments without prior notice.

Please reach out to me directly if you would like to request accommodations for the course to better facilitate your learning. Student Disability Services (disabilities.uchicago.edu) is

also available to provide resources and support, and may provide approval for specific academic accommodations. Informing me in a timely manner will help me ensure accommodations are met and I am able to implement an appropriate assessment of your learning.